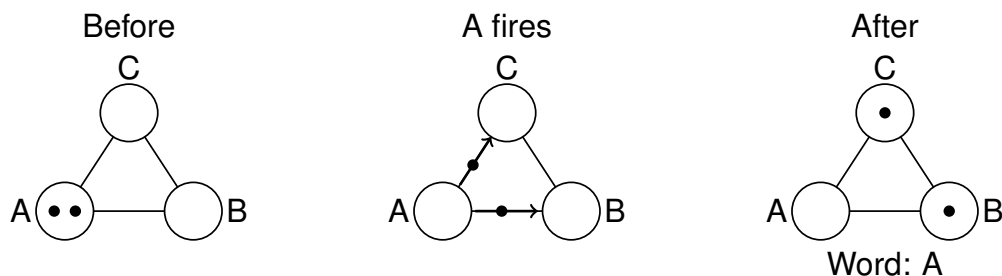
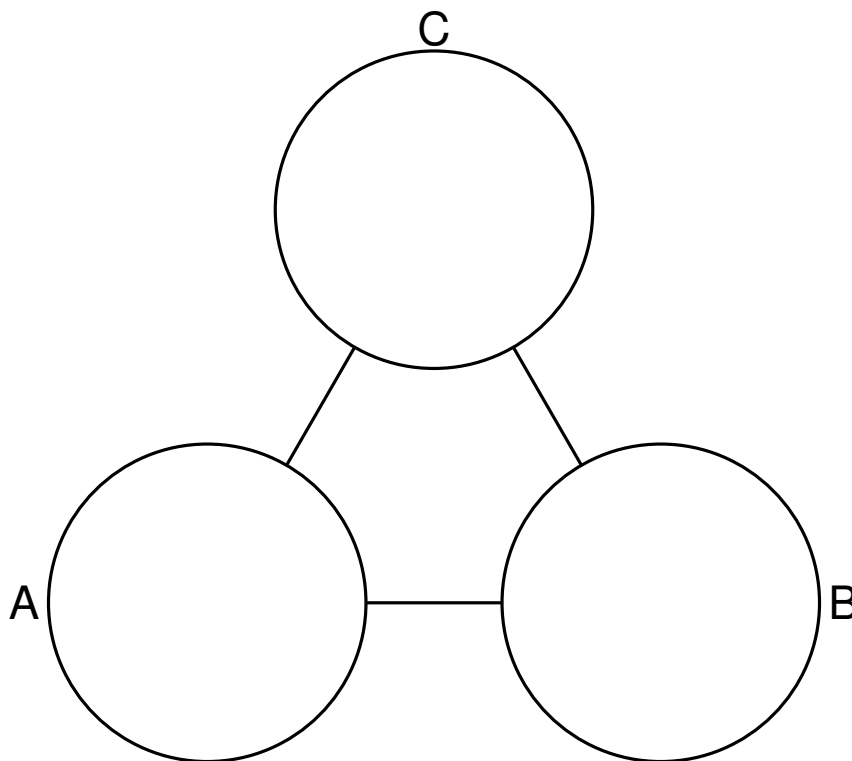


Choose one circle at a time. It can fire when it has at least one chip for each line from it. Send one chip from it along each line. A sink never fires; chips there stay. Write the firing circle's letter after each move.



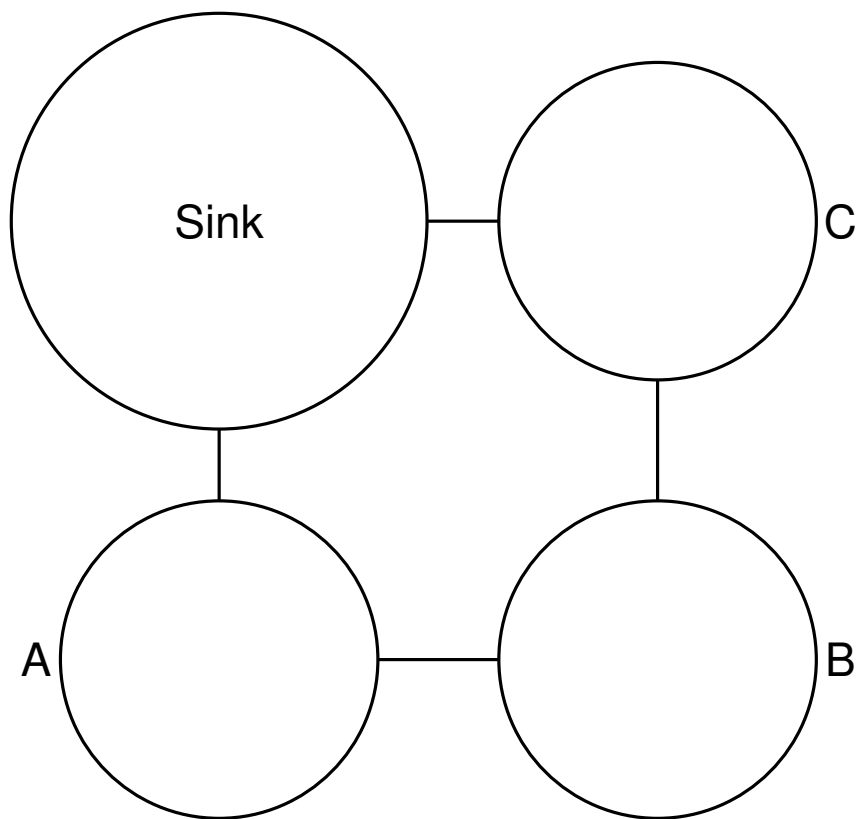
Problem 1: Try these starts. Which can stop with no circle able to fire? Which can return to an earlier position?

Start 1	Start 2	Start 3
A: 3, B: 0, C: 0	A: 2, B: 1, C: 0	A: 4, B: 0, C: 0



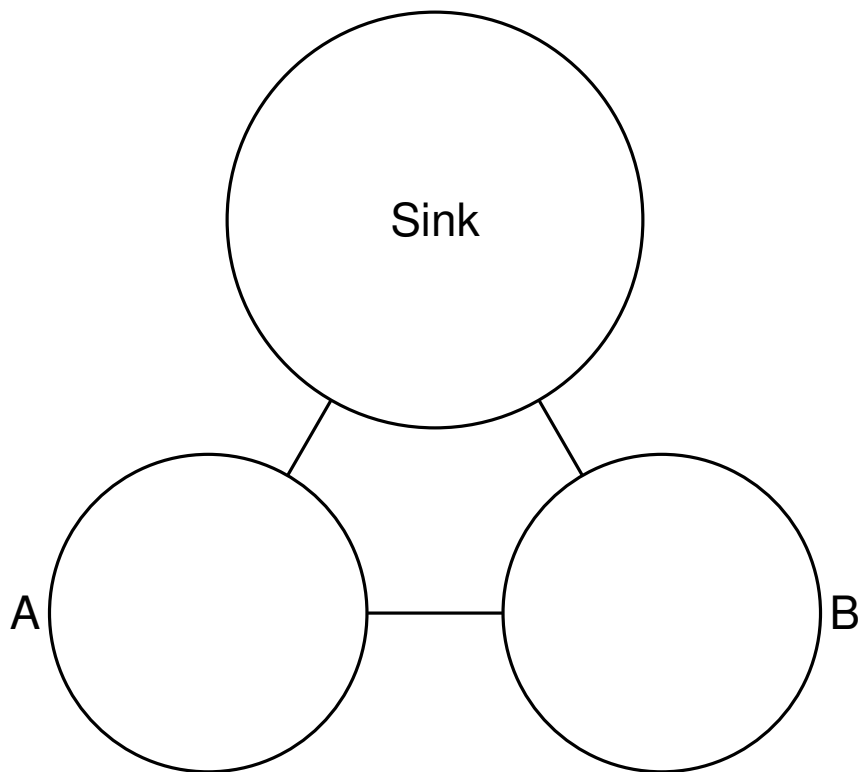
Start	Firing letters, in order	Finish or repeat
1		
2		
3		

Problem 2: Put one chip at each of A, B and C. Add one chip at A, B or C, then fire until no circle can fire. For each trial, restore the chosen starting piles and empty the sink before adding a chip. Among all starts with at most one chip at each of A, B and C, which start and added chip make the longest firing word?



Start A, B, C	Add at	Firing letters, in order	Finish A, B, C
1, 1, 1	A		
1, 1, 1	B		
1, 1, 1	C		

Problem 3: Each start below can finish with one chip at A, one at B and four in the sink, using exactly four firings. The sink starts empty. Find every legal four-letter firing word for each start.



Start: A, B	Legal four-letter firing words
0, 6	
3, 3	
6, 0	